

Weighted Automata Learning with a Hard Inductive Bias

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Main Idea

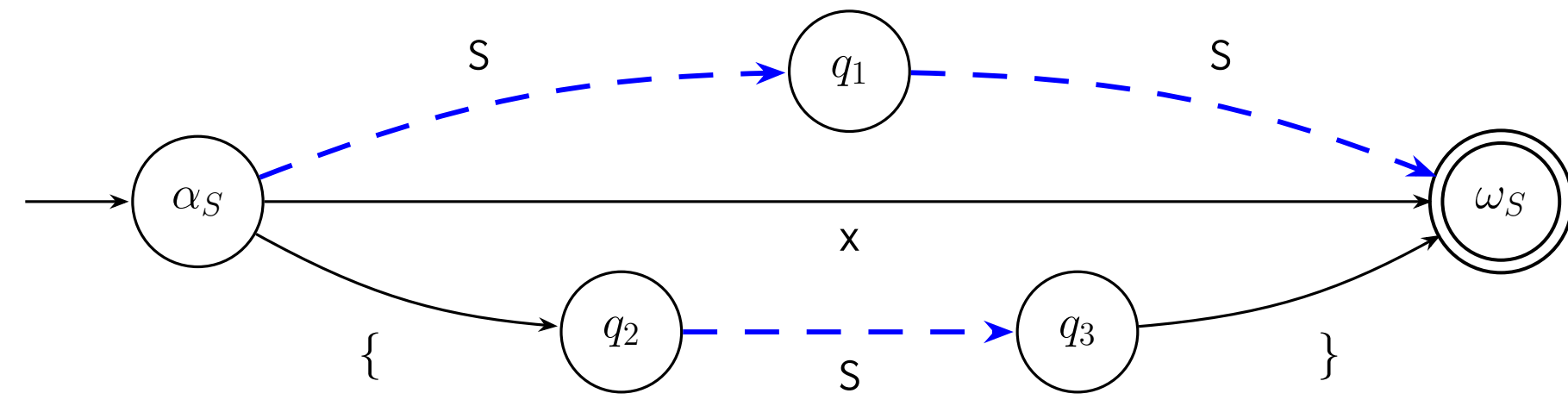
- **Assume:** we are given a learned distribution, $P_\theta(\sigma)$, with support over $\sigma : \Sigma^+$ but largely concentrated within $\mathcal{L}(G)$, where G is a known CFG.
- **Our goal:** distill a surrogate WFA, α_θ , such that for all $\sigma \in \mathcal{L}(G)$, $P_\theta(\sigma) \propto P_\theta(\sigma \mid \sigma \in \mathcal{L}(G))$, while preserving inclusion ($\mathcal{L}(\alpha) \supseteq \mathcal{L}(G)$).
- **Desiderata:** The regular description α should be reasonably concise and precise, i.e., $\Delta = |\mathcal{L}(\alpha) \setminus \mathcal{L}(G)|$ is approximately minimized up to Σ^n .

Nederhof's Construction

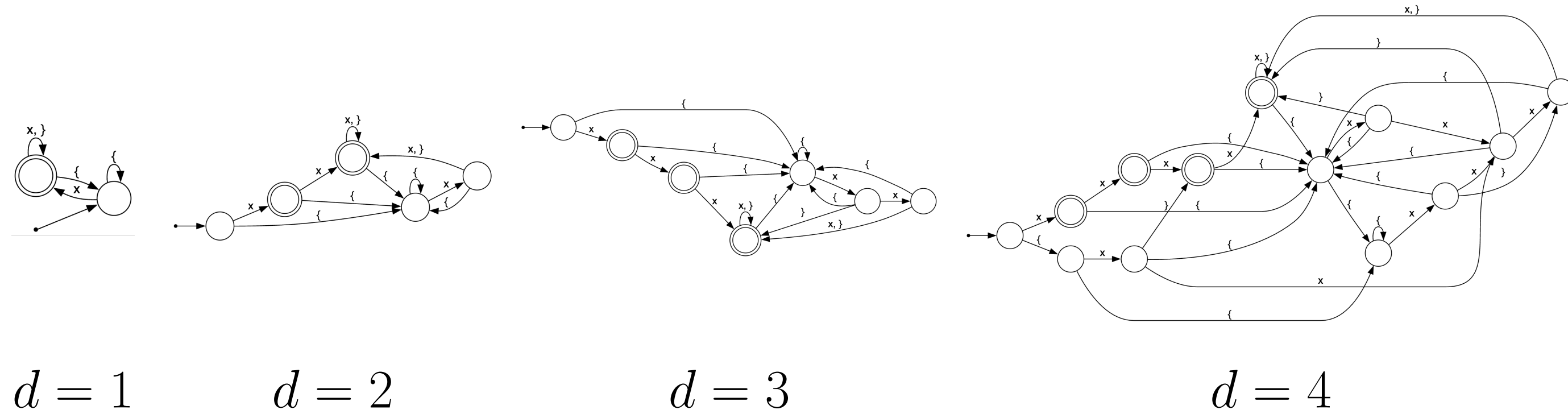
A recursive transition network (RTN) of a CFG $\langle \Sigma, V, P, S \rangle$ is an automaton,

$$\mathcal{R} = \{R_A = (Q_A, \delta_A, \alpha_A, F_A)\}_{A \in V}, \quad \delta_A \subseteq Q_A \times (\Sigma \cup V \cup \{\varepsilon\}) \times Q_A,$$

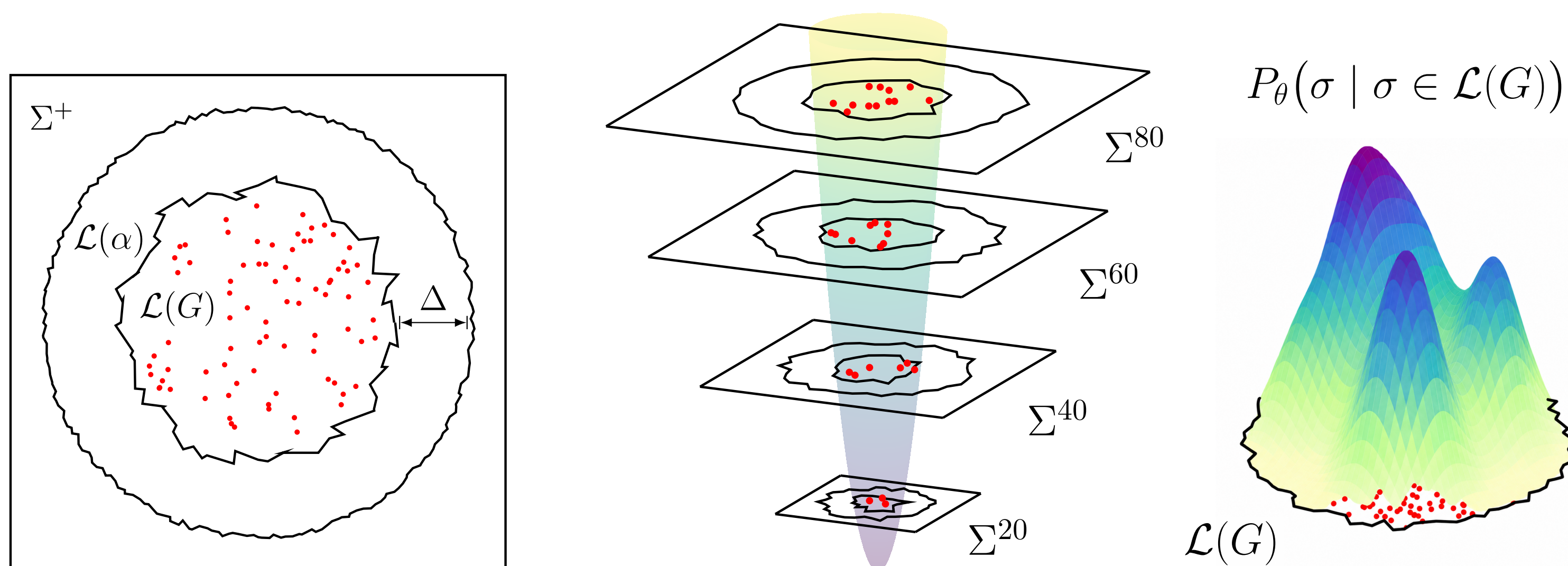
where an arc labelled $B \in V$ denotes a call to R_B . For example, consider the CFG, $S \rightarrow x \mid S S \mid \{ S \}$, which has the following RTN:



Nederhof's d -RTN approximation is a depth-bounded RTN unfolding. A nonterminal arc $p \xrightarrow{B} q$ is replaced by a fresh copy of the RTN for B , adding ε -arcs from p to the copy's initial state and the copy's final state back to q : $p \xrightarrow{\varepsilon} \alpha'_B$, $\omega'_B \xrightarrow{\varepsilon} q$ ($\forall \omega'_B \in F'_B$). Unfold calls for d rounds. At depth d , every remaining nonterminal arc $p \xrightarrow{B} q$ is woven into a shared copy of R_B : $p \xrightarrow{\varepsilon} \alpha_B$, $\omega_B \xrightarrow{\varepsilon} q$ ($\forall \omega_B \in F_B$). Below is the ε -removed, determinized and minimized DFA depicted for $1 \leq d \leq 4$:



Language Overapproximation



We can measure over-approximation tightness by considering finite models,

$$\rho(\alpha, G, m) = \frac{\sum_{n=1}^m |\mathcal{L}(G) \cap \Sigma^n|}{\sum_{n=1}^m |\mathcal{L}(\alpha) \cap \Sigma^n|}, \quad \alpha \sqsubseteq_G^m \alpha' \iff \rho(\alpha', G, m) < \rho(\alpha, G, m).$$

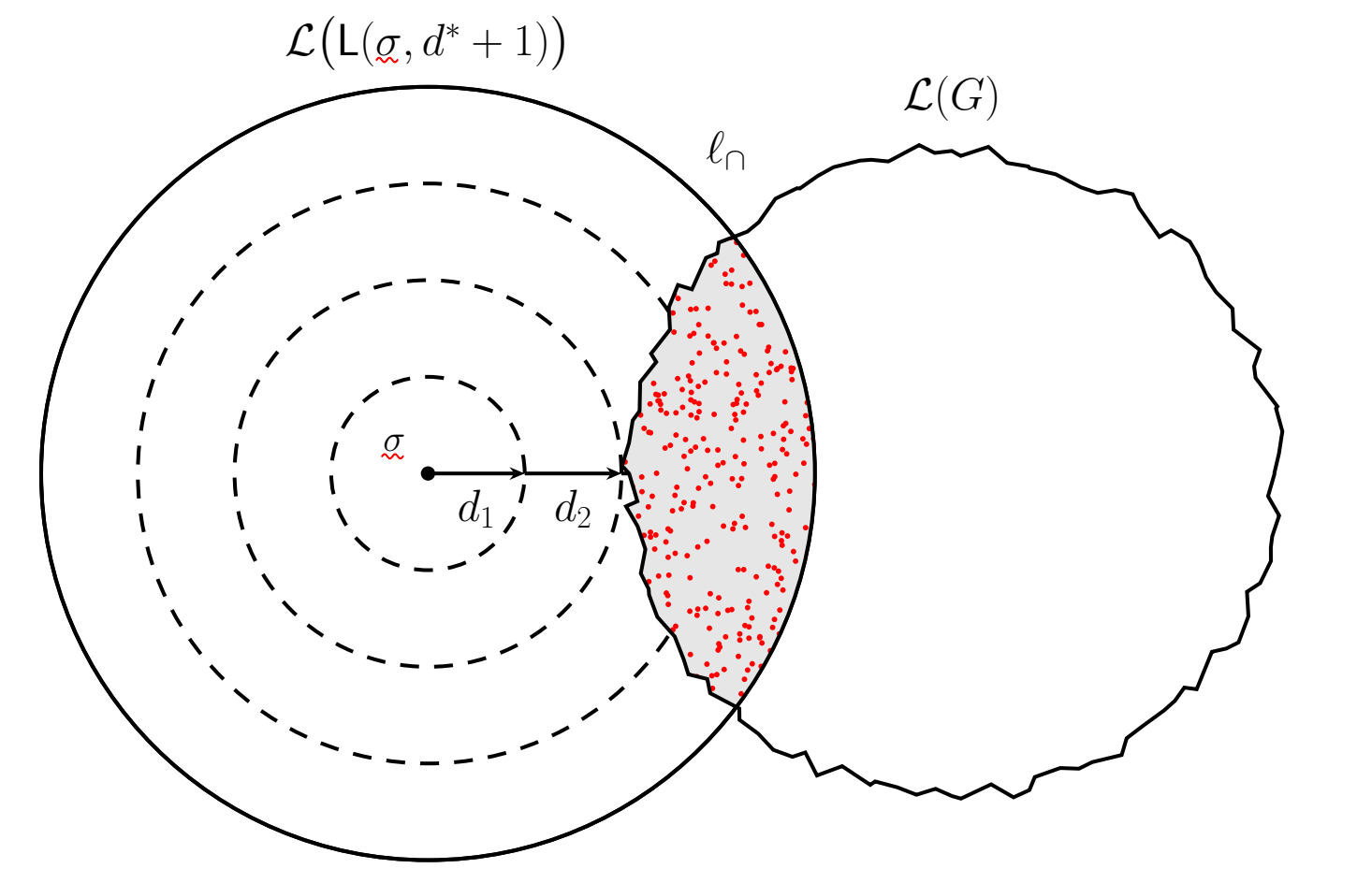
Method

We propose a counterexample-guided abstraction refinement (CEGAR) loop that incrementally refines α_θ until a desired precision or $|Q|_{\max}$ is reached:

1. Initialize $\alpha'_\theta \leftarrow \alpha_d$ using Nederhof's d -RTN construction.
2. Propose a refinement $D_{\text{KL}}(P_\theta \| \alpha'_\theta) < D_{\text{KL}}(P_\theta \| \alpha_\theta)$ and $\alpha_\theta \sqsubseteq_G^m \alpha'_\theta$.
3. Binarize α_θ and verify $\mathcal{L}(\alpha) \supset \mathcal{L}(G)$ by checking $\emptyset = \mathcal{L}(G) \cap \overline{\mathcal{L}(\alpha)}$.
4. If verification fails, collect $x \in \mathcal{L}(G) \cap \overline{\mathcal{L}(\alpha)}$, otherwise go to step 2.
5. Repair α restoring inclusion, renormalize as α'_θ , then go to step 2.

Intersection Decoding

We will evaluate our learned WFA on a constrained decoding task: given a dataset of near-words $\sigma \notin \mathcal{L}(G)$ and pairwise human repairs σ' within a small edit distance, sample uniformly WoR repairs within $\ell_\cap$, score and rank each repair according to $P_\theta(\sigma)$, then measure the true repair's rank.



Definition 1 (Bounded Levenshtein-CFL reachability). Given a CFL, ℓ , and an invalid string, $\sigma : \bar{\ell}$, find every valid string reachable within d edits of σ , i.e., if Δ_L is the Levenshtein metric and $\mathcal{L}(L(\sigma, d)) = \{\sigma' \mid \Delta_L(\sigma, \sigma') \leq d\}$ is the Levenshtein d -ball, we seek to find $\ell_\cap = \mathcal{L}(L(\sigma, d)) \cap \ell$.

Moreover, we want a procedure that surfaces both natural and valid repairs:

Definition 2 (Ranked repair). Given a finite language $\ell_\cap = \mathcal{L}(L(\sigma, d)) \cap \ell$ and LM, $P_\theta : \Sigma^* \rightarrow \mathbb{R}$, find the top- k maximum probability repairs, i.e.,

$$R(\ell_\cap, P_\theta) : 2^{\Sigma^+} \times (\Sigma^+ \rightarrow \mathbb{R}) \rightarrow (\Sigma^+)^{\leq k} = \arg \max_{\sigma \subseteq \ell_\cap, |\sigma| \leq k} \sum_{\sigma \in \sigma} P_\theta(\sigma)$$

Evaluation

We will compare four different methods for constrained decoding from $\ell_\cap$:

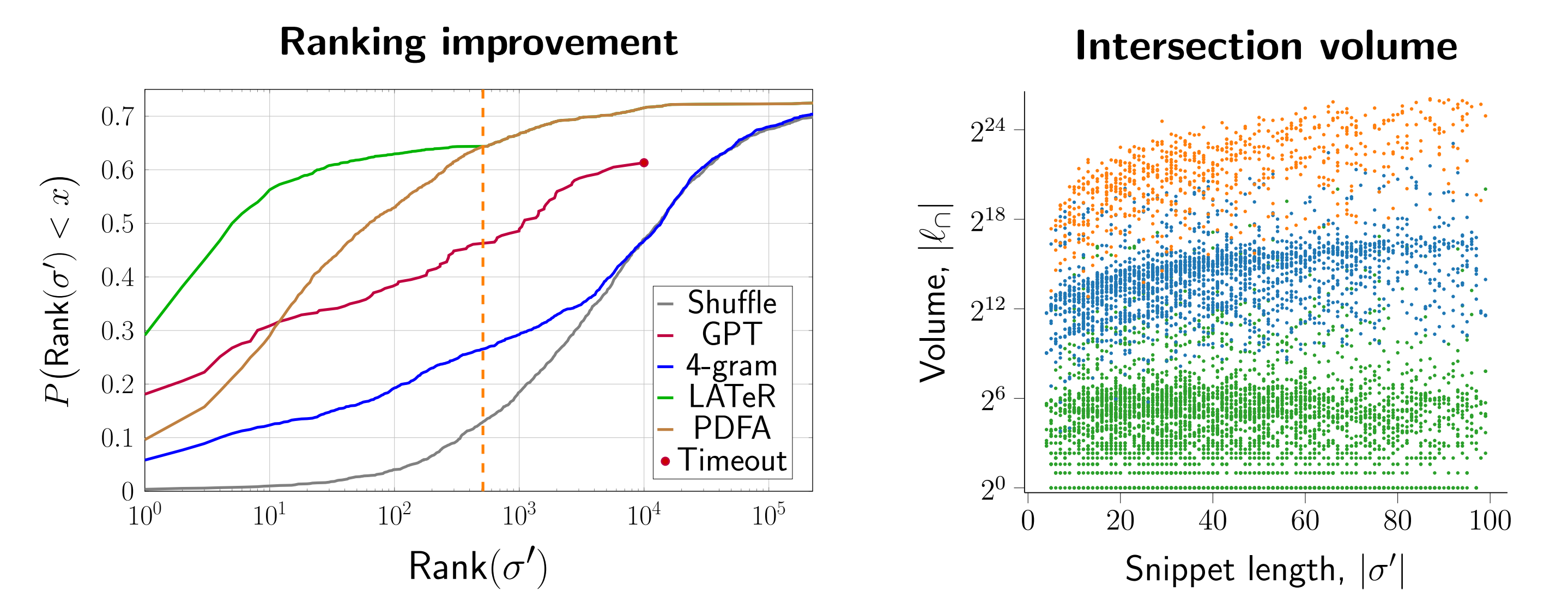
1. A **4-gram** model with Laplace smoothing trained on $\sigma \in \mathcal{L}(G)$
2. A **GPT**-style model trained on $\sigma \sigma'$ with LTR σ' masking
3. A **PDFA** distilled from P_θ overapproximating $\mathcal{L}(G)$
4. A **LATeR** (PDFA decoding + top-512 reranker)

Let $E_\theta : \Sigma^+ \rightarrow \mathbb{R}^p$ be a transformer encoder and $E'_\theta : \Sigma^+ \times \mathbb{Z}_4^+ \rightarrow \mathbb{R}^p$ be an encoder which takes in a Levenshtein alignment ($\text{LA} : \Sigma^+ \times \Sigma^+ \rightarrow \mathbb{Z}_4^+$) for each $\langle \sigma, \sigma' \rangle$ pair. Now add an MLP and output a scalar relevance score:

$$\Delta_\theta(\sigma, \sigma') = \text{MLP}_\theta(E_\theta(\sigma), E'_\theta(\sigma', \text{LA}(\sigma, \sigma')))$$

We call this model a Levenshtein-Aligned Transformer Reranker (LATeR) and train on $\langle \sigma, \sigma' \rangle$ with PDFA-sampled confounders and a listwise CE loss.

Results



We depict cumulative rank and intersection volume across the test set.

Conclusion

- Single-threaded PDFA sampling exhibits $10^3 - 10^5 \times$ higher throughput than constrained LLM decoding, often saturating $\ell_\cap$ in under 100ms.
- The size- k subset maximizing P_θ over $\ell_\cap$ can far outweigh the mass from an LLM-guided subsampling. We call this the **decoder-retriever gap**.
- Hence, exhaustive ranking with a low-fidelity but fast P_θ approximation is preferable to decoding a highly calibrated but low-throughput LLM.
- Is there a succinct construction (possibly with a very different topology) that soundly overapproximates $\mathcal{L}(G)$ and more faithfully reconstructs P_θ ?